SOMAIYA VIDYAVIHAR UNIVERSITY

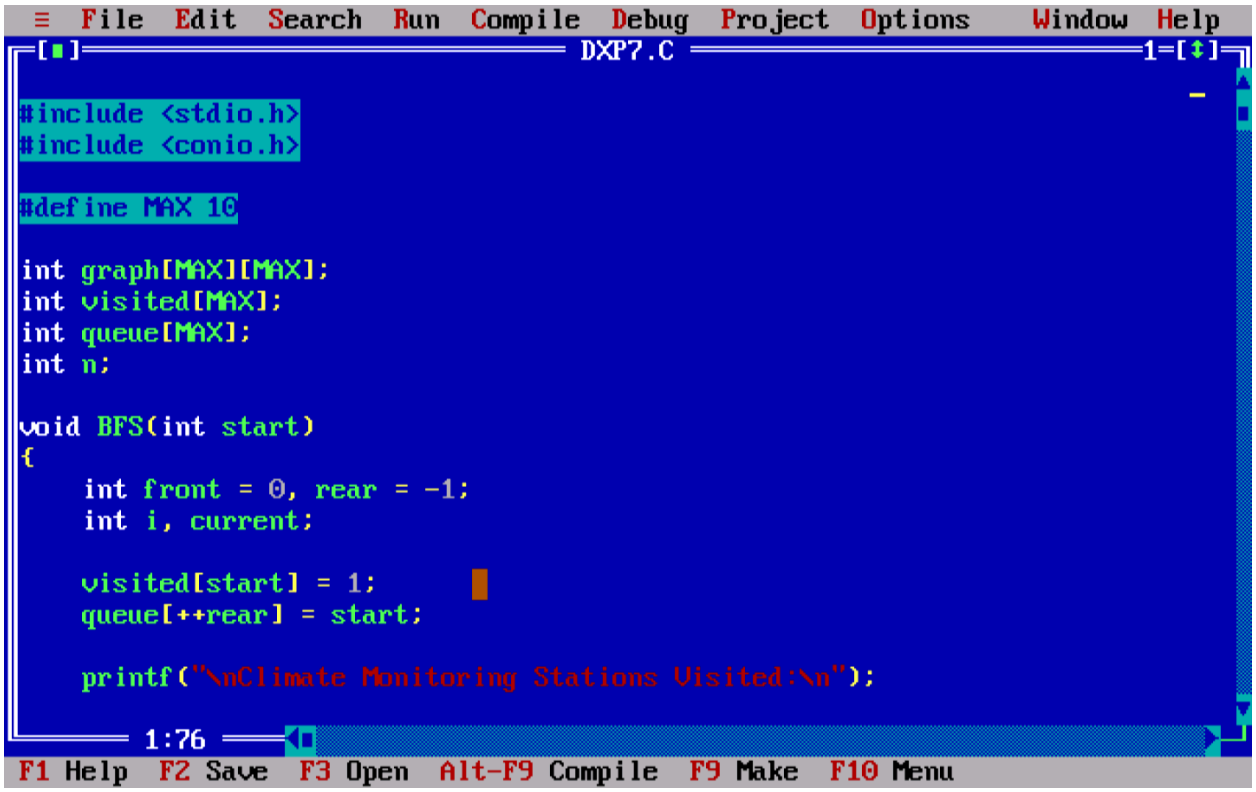
K J Somaiya Institute of Technology

An Autonomous Institute Permanently Affiliated to the University of Mumbai

DEPARTMENT OF INFORMATION TECHNOLOGY

Name of Student: Soham Dagadu Kathe
Roll No : B - 68
Batch : SB4

Program:



```
#include <stdio.h>
#include <conio.h>

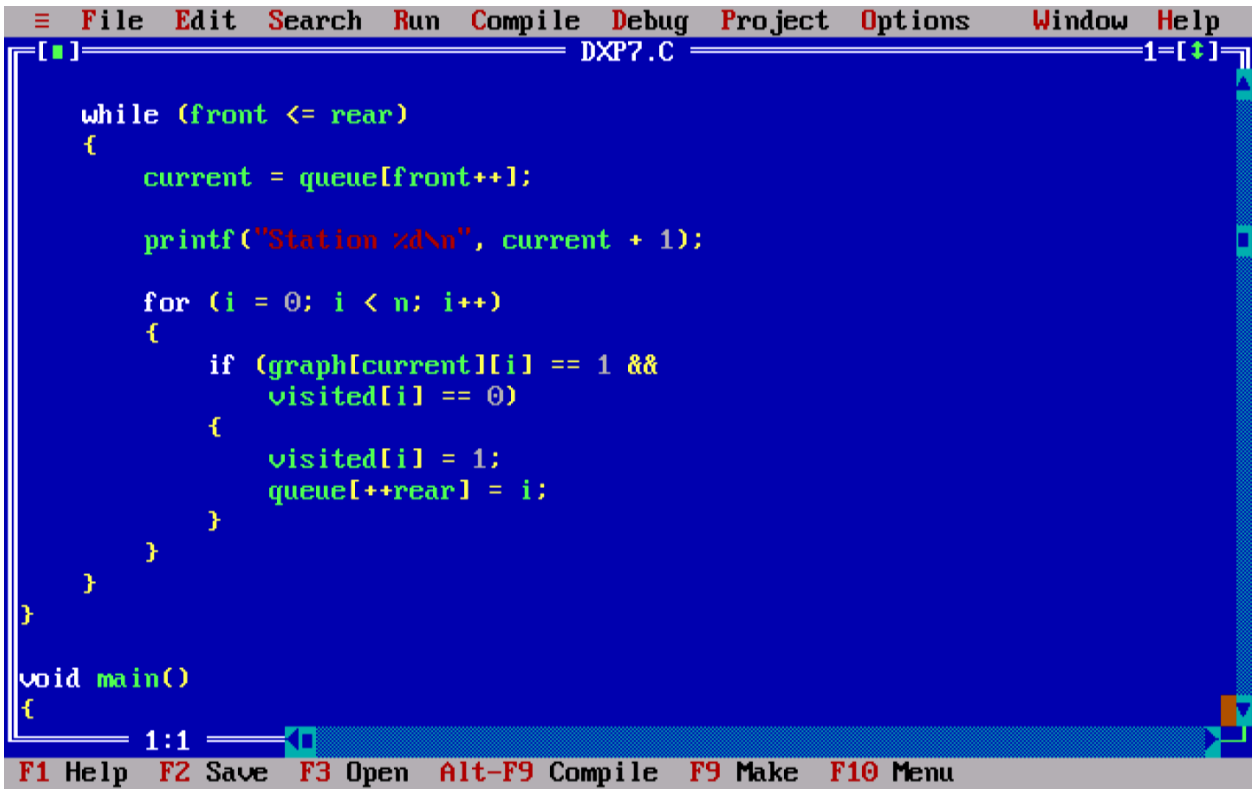
#define MAX 10

int graph[MAX][MAX];
int visited[MAX];
int queue[MAX];
int n;

void BFS(int start)
{
    int front = 0, rear = -1;
    int i, current;

    visited[start] = 1;
    queue[++rear] = start;

    printf("\nClimate Monitoring Stations Visited:\n");
```

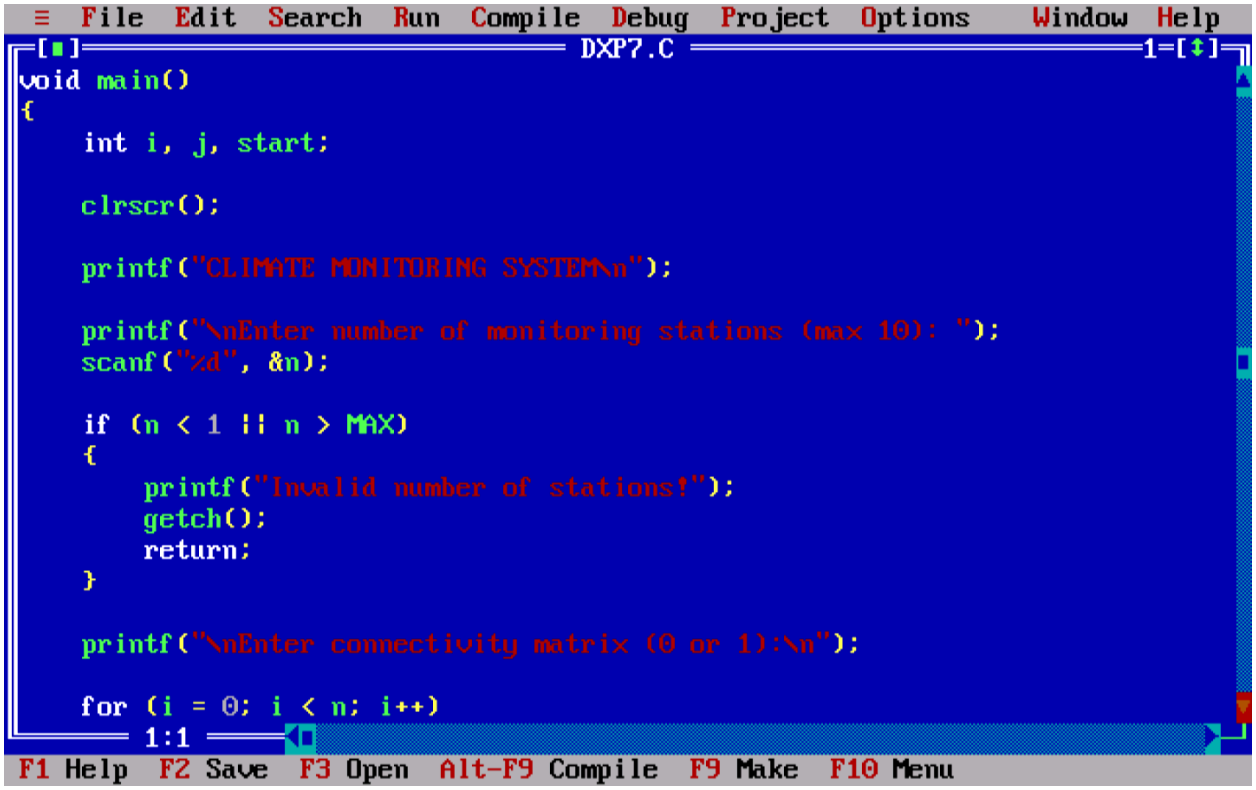



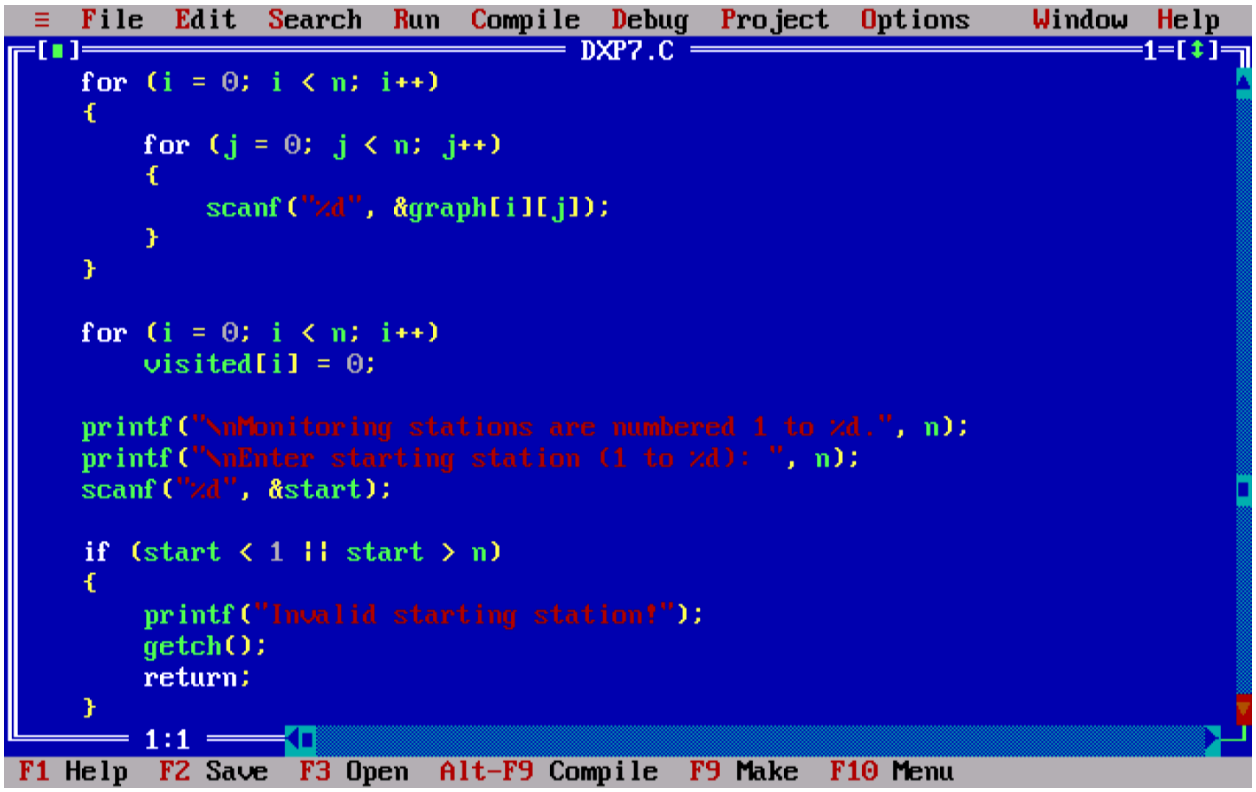
```
    while (front <= rear)
    {
        current = queue[front++];

        printf("Station %d\n", current + 1);

        for (i = 0; i < n; i++)
        {
            if (graph[current][i] == 1 &&
                visited[i] == 0)
            {
                visited[i] = 1;
                queue[++rear] = i;
            }
        }
    }
}

void main()
{
```


File Edit Search Run Compile Debug Project Options Window Help
DXP7.C
1=[↕]
void main()
{
int i, j, start;
clrscr();
printf("CLIMATE MONITORING SYSTEM\n");
printf("\nEnter number of monitoring stations (max 10): ");
scanf("%d", &n);
if (n < 1 || n > MAX)
{
printf("Invalid number of stations!");
getch();
return;
}
printf("\nEnter connectivity matrix (0 or 1):\n");
for (i = 0; i < n; i++)
1:1
F1 Help F2 Save F3 Open Alt-F9 Compile F9 Make F10 Menu


File Edit Search Run Compile Debug Project Options Window Help
DXP7.C
1=[↕]
for (i = 0; i < n; i++)
{
for (j = 0; j < n; j++)
{
scanf("%d", &graph[i][j]);
}
}
for (i = 0; i < n; i++)
visited[i] = 0;
printf("\nMonitoring stations are numbered 1 to %d.", n);
printf("\nEnter starting station (1 to %d): ", n);
scanf("%d", &start);
if (start < 1 || start > n)
{
printf("Invalid starting station!");
getch();
return;
}
1:1
F1 Help F2 Save F3 Open Alt-F9 Compile F9 Make F10 Menu




```
CLIMATE MONITORING SYSTEM

Enter number of monitoring stations (max 10): 4

Enter connectivity matrix (0 or 1):
0 1 1 0
1 0 0 1
1 0 0 1
0 1 1 0

Monitoring stations are numbered 1 to 4.
Enter starting station (1 to 4): 1

Climate Monitoring Stations Visited:
Station 1
Station 2
Station 3
Station 4
```